Sevilla, Spain Feb 10-12, 2016

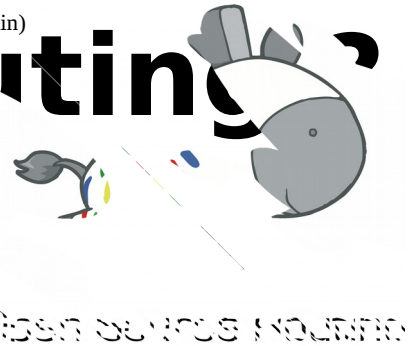
Challenges in Testing

How OpenSourceRouting tests Quagga

Martin Winter
Feb 10, 2016

Open Source Routing

Who is OpenSourceRouting?



- **Who is Open Source Routing ?**
  - www.opensourcerouting.org
  - Project by NetDEF (Network Device Education Foundation)
    - www.netdef.org
    - Non-Profit Company based in California
  - Working on Quagga Routing

netDEF
The Network Device Education Foundation

- **Who is Martin Winter ?**
  - Co-Founder of NetDEF
  - Focusing on Testing Quagga
  - Previously worked for Equipment Vendor & large ISP

What is Quagga ?

- **Routing Protocol Stack**
  - RIP / RIPNG / OSPFv2 / OSPFv3 / ISIS / BGP / PIM
  - Running on Linux / FreeBSD / NetBSD / OpenBSD / Solaris
  - Used on low-end OpenWRT boxes, physical and virtual software routers, SDN deployments, distributed routers
  - Originally derived from Zebra
  - GPLv2+ Open Source / “Community” owned & controlled

Quagga Community

How it works today

No "Owner"

No single entity behind Quagga

Large community of "contributers"

Maintainer = person with commit access

Simple Git Model

Main source git on Savannah

Single master branch with development branch merged into every few months

Email based submissions

Code submissions by email to developer list

Code review with discussion on list

Choosing a CI System

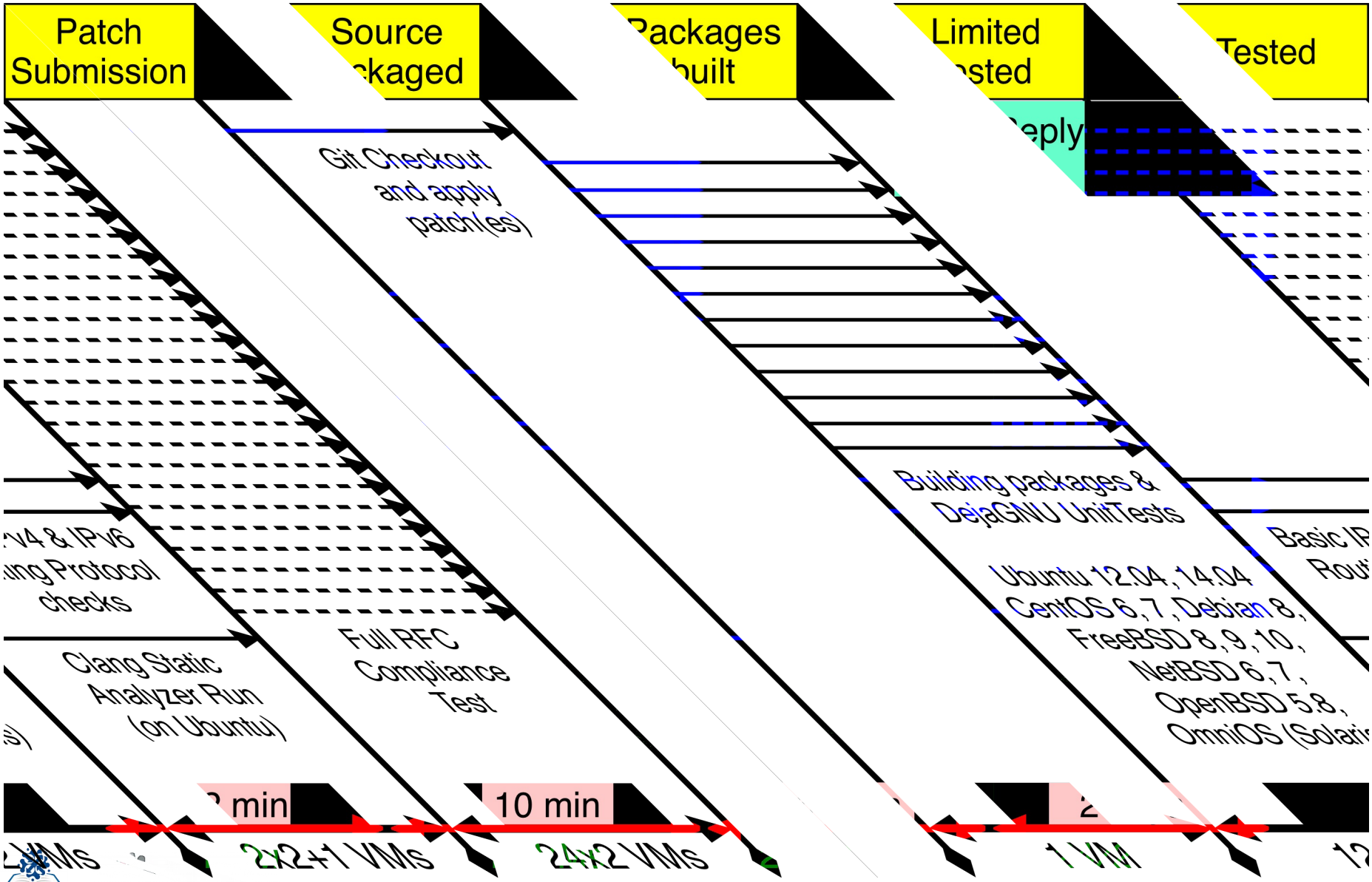
Jenkins

Travis CI

- **Are all CI systems designed for Linux & Java?**
  - Java clients on agents are NOT cross-platform
  - There are other OS'es than Linux in existence
- **Stability**
- **Hosted vs Local**
  - Integrate Dedicated Tester Hardware?
  - Limitations on Parallelization / Runtime?

CI Run Overview

Open Source Routing



Git Checkout / Patching

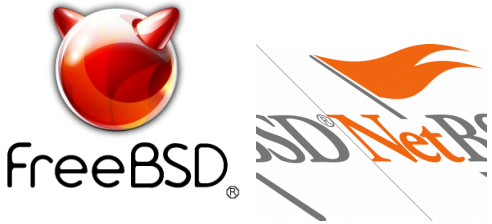
All starts with new commit submitted

- **Pull code from Git**
  - Which revision / branch?
- **Add patches**
  - Single patch / Series of patches?
- **Package Source**
  - Add Extra Information (Git Rev, Patches, build time etc)

Building Packages on each OS

Running on parallel VMs

- **Running on VMs – start/shut on der**
- **Reset VM to clean snapshot at beginning**
- **Configure the code**
  - ./configure – but which choices?
- **Build (make & make install)**
- **Run Unittests (DejaGNU [make cl**
- **Build OS specific packages**
  - Different configuration settings? All feat



Basic Protocol checks

- **Use RFC compliance checker for a 2..3 selected checks on each protocol**
  - Just make sure they don’t immediately crash
  - Parallel/Serialize? Balance between runtime and required resources (multiple CI runs in parallel possible!)
- **Rebuild with CLANG**
  - Use Clang Static Analyzer
  - How to translate results into pass/fail ? Diff to previous run?

Automated reply

- **Sent as reply to Patch em**
- **Reply approx 1..2 hrs after submiss**
  - Earlier if failing at beginning of tests
- **Parse results into easy email**
  - Assume submitter is first-timer and doesn't know tests
  - Simplify results

Example: Successful submission

From: cisystem@netdef.org
To: [redacted]
Cc: mwinter@opensourcerouting.org, quagga-dev@lists.quagga.net
Subject: CI Testresult: PASSED (Re: [quagga-dev,14658] config: Remove unused library check)
Date: February 2, 2016 at 7:40 PM

Continous Integration Result: SUCCESSFUL

Congratulations, this patch passed basic tests

Tested-by: NetDEF CI System <cisystem@netdef.org>

This is an EXPERIMENTAL automated CI system.
For questions and feedback, feel free to email
Martin Winter <mwinter@opensourcerouting.org>.

Patches applied :
Patchwork 1811: http://patchwork.quagga.net/patch/1811
[quagga-dev,14658] config: Remove unused library check
Tested on top of Git : eae18d1 (as of 20151209.135437 UTC)
CI System Testrun URL: https://ci1.netdef.org/browse/QUAGGA-QPWORK-231/

Regards,
NetDEF/OpenSourceRouting Continous Integration (CI) System

OpenSourceRouting.org is a project of the Network Device Education Foundation,
For more information, see www.netdef.org and www.opensourcerouting.org
For questions in regards to this CI System, contact Martin Winter, mwinter@netdef.org

From: cisystem@netdef.org
To: timo.teras@iki.fi
Cc: mwinter@opensourcerouting.org, quagga-dev@lists.quagga.net
Subject: CI Testresult: FAILED (Re: [quagga-dev,14376,v3] lib, zebra: unify link layer type and hardware address handling)
Date: December 26, 2015 at 2:20 AM

Continous Integration Result: FAILED

See below for issues.
This is an EXPERIMENTAL automated CI system.
For questions and feedback, feel free to email
Martin Winter <mwinter@opensourcerouting.org>.

Patches applied :
Patchwork 1741: http://patchwork.quagga.net/patch/1741
[quagga-dev,14376,v3] lib, zebra: unify link layer type and hardware address handling
Tested on top of Git : eae18d1 (as of 20151209.135437 UTC)
CI System Testrun URL: https://ci1.netdef.org/browse/QUAGGA-QPWORK-204/

Get source and apply patch from patchwork: Successful

Building Stage: Failed

CentOS7 amd64 build: Successful
Debian8 amd64 build: Successful
Ubuntu1204 amd64 build: Successful
CentOS6 amd64 build: Successful
Ubuntu1404 amd64 build: Successful

Make failed for FreeBSD10 amd64 build: (see full log in attachment freebsd10_amd64_make.log)

```
  CC      rtread_sysctl.o
  CC      kernel_socket.o
kernel_socket.c:1127:41: error: no member named 'sdl' in 'struct interface'
     gate = (union sockunion *) & ifp->sdl;
                                  ~~~  ^
1 error generated.
*** Error code 1
Stop.
make[2]: stopped in /usr/home/ci/cibuild.204/quagga-source/zebra
```

Make failed for NetBSD6 amd64 build: (see full log in attachment netbsd6_amd64_make.log)

```
  CC      kernel_socket.o
```

Example: Failed compilation

Example: Failed Basic Routing Checks

Continous Integration Result: FAILED

See below for issues.
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Patches applied :
Patchwork 1808: http://patchwork.quagga.net/patch/1808
[quagga-dev,14647] perfect cmd table
Tested on top of Git : eae18d1 (as of 20151209.135437 UTC)
CI System Testrun URL: https://ci1.netdef.org/browse/QUAGGA-QPWORK-228/

Get source and apply patch from patchwork: Successful

Building Stage: Successful

Basic Tests: Failed

Ipv6 protocols: Successful
Static analyzer (clang): Successful

RFC Compliance Test ANVL-BGP4-4.1 failing:

> Test Summary
> An UPDATE message MAY simultaneously advertise a feasible route and withdraw multiple unfeasible routes from service.
> Test Reference
> RFC4271, Sect. 4.3, p 15,
> UPDATE Message Format
> Test Classification
> MAY
> Test ANVL-BGP4-4.1: !FAILED!
> Received unexpected IP packet
> External peer <PEER-AS2-A> incorrectly received IP packet forwarded by DUT

Run Full RFC Compliance Check

- **More than just compliance**
  - Heavy interaction with CLI -> Good CLI Testing as well

ixia IxANVL

- **Requires between 3 and 30 hours per protocol**
- **Multiple runs required to find inconsistent results**
- **Feed all results, logs, pcaps into DB**
- **Create PDF report**
- **Run against commercial/other products to compare**
  - (but not publishing results from commercial equipment)

see https://www.opensourcerouting.org/compliance-test-results/

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Open Source Routing

Protocol Fuzzer

- **Hardware – 4 parallel tests only**
  - No open source solution available
- **Slow. Single pass of BGP takes ~1..2 weeks runtime**
  - For single configuration, when no errors are found
- **How to publish results?**
  - All pass: boring
  - Some fail: Security issue, open CVE first?

Scale/Performance tests

- Hard to find a consistent way to compare
  - Anyone having success running performance/scale tests on VMs?
- **Expensive Equipment (+ Power + Automation)**
- **Spin-up/down on demand and automatically building topology with “patchpanels” is painful**
  - Ie Using old Cisco 4948 for patchpanel (cheaper than OpenFlow switches)
- **Using Ixia IxNetwork and Spirer TestCenter**
- **Pass/Fail criteria?**

Contact Us

Questions?

Comments?

Want to discuss?

Want to Sponsor?

Martin Winter **mwinter@netdef.org**

Network Device Education Foundation (NetDEF)

www.netdef.org /

www.opensourcerouting.org